

SCORIFY Phase 2

Dataset Source & Description

We got our dataset through a personal connection. One of our close team members is currently working at a Netherlands-based company that partners with Go2Africa, which is a well-known luxury African safari travel company that has been operating for over 25 years. We reached out to him and asked if he could help us get some real lead data for our project. He coordinated with the team there and shared an anonymized data export from their CRM system (HubSpot), pulled through BigQuery.

The raw file has 3,931 rows and 224 columns. Each row is one lead — basically a person who showed interest in booking a safari trip. The columns cover all sorts of things: where the lead came from (Google ads, Facebook, organic search, direct), what device they used, which country they are from, what destinations they looked at, their budget, how many pages they viewed, how long they spent on the website, whether they filled out forms, and so on. There are also some internal sales pipeline fields like deal stages and activity counts.

For our Scorify project we are using two columns as prediction targets: the conversion probability (a value between 0 and 1 showing how likely a lead is to convert) and the predicted profit (how much profit the company expects if that lead actually books). These are the two things our models will learn to predict.

Data Collection Method

The data came directly from Go2Africa's operations through our contact at the Netherlands-based company. It was not scraped or taken from any public source — it is actual business data from a real company with real leads. The export was done from their HubSpot CRM via BigQuery, which is how they store and manage their lead records internally.

Before sharing the data with us, all personal information was stripped out. There are no names, emails, phone numbers, or anything that could identify an individual lead. What we received is purely behavioral and profile data — things like how many times someone visited the website, what pages they looked at, their stated budget range, which marketing channel brought them in, and whether the sales team followed up.

The dataset covers leads from multiple acquisition channels including paid social media (Facebook, Instagram), paid search (Google), organic

search, direct traffic, email campaigns, and referrals. This gives us a good mix of lead types to work with for training our scoring model.

Preprocessing Steps

The raw data needed quite a bit of cleaning before it was usable for model training. Here is what we did step by step:

- **Deduplication:** We noticed the same lead (same record id) appeared multiple times in the data because they get re-scored over time. We kept only the most recent entry for each lead, which brought us from 3,931 rows down to 2,458
- **Column selection:** Out of 224 columns, most were not useful for our purpose. We dropped things like email addresses, click IDs, Google Analytics IDs, and other identifiers that have zero predictive value. We also removed all timestamp columns, system flags (like `updated_hubspot`, `has_been_scored`), empty columns (around 67 Feature columns had literally 0% data), free-text message fields, and duplicate columns (for example `Feature8` and `custom_prop1` had identical values, same with `Feature1` and `custom_prop2`). After all this we were left with 43 columns.
- **Renaming:** A lot of the columns had generic names like `Feature6`, `Feature48`, `custom_prop2` etc. which made no sense on their own. We discussed the data with one of the company employees to understand what each field actually means. Based on that conversation, we renamed columns to meaningful labels — for example `Feature6` became `enquiry_touchpoints`, `Feature48` became `budget_level`, `custom_prop1` became `travel_group_type`, and so on. This makes the dataset much easier to work with during model training.
- **Missing value imputation:** For numeric columns we filled missing values with the median of that column. For categorical columns we filled blanks with 'Unknown'. This way we do not lose any rows.
- **95% null/zero filter:** After imputation, we checked each column to see if it was mostly empty or zero. Six columns had 95% or more of their values as zero: `returning_visitor` (99.7%), `contact_count` (98.2%), `has_specific_dates` (97.4%), `follow_up_count` (96.7%), `offer_count` (96.8%), and `response_indicator` (95.2%). These columns carry almost no useful information so we dropped them.
- **Zero-variance check:** We also checked for columns where every row has the same value — none were found after the previous filtering steps.

After all these steps our final cleaned dataset has 2,458 rows and 37 columns — 1 ID column (`record_id`), 34 input features, and 2 target variables. There are zero missing values.

4. EDA Key Insights

We ran some basic exploratory analysis to understand what we are working with before jumping into model building.

Target variables:

The conversion probability is heavily skewed to the right. Most leads sit below 5% probability, and only a handful go above 20%. The mean is 0.051 and median is just 0.019 which honestly makes sense because in real life most people who visit a travel website are just browsing, not ready to book. The profit target ranges from about 26K to 670K USD, with a mean around 112K. Higher-end safari packages push the upper range.

What features matter most:

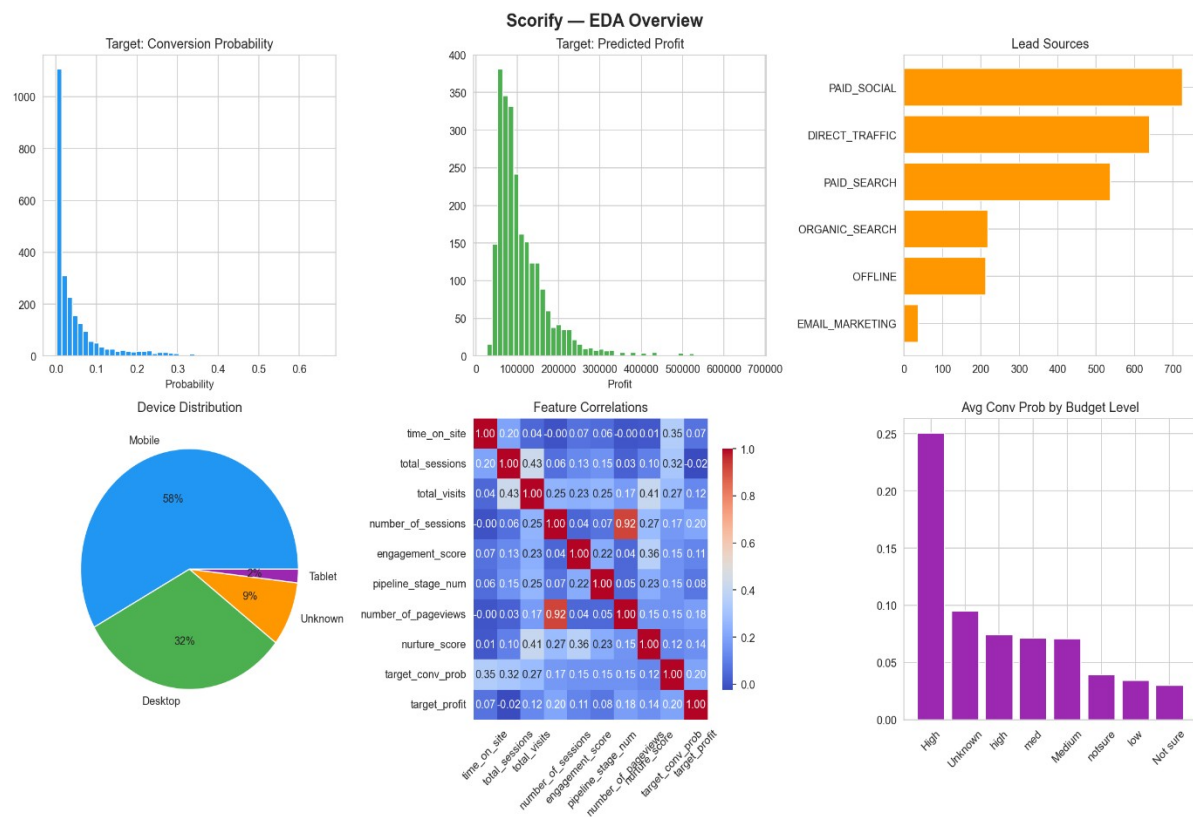
For Conversion Probability	Corr	For Profit Prediction	Corr
time_on_site	0.35	enquiry_touchpoints	0.61
total_sessions	0.32	page_interactions	0.57
total_visits	0.27	number_of_sessions	0.20
number_of_sessions	0.17	number_of_pageviews	0.18
engagement_score	0.15	nurture_score	0.14

The biggest takeaway is that behavioral features dominate. How much time someone spends on the site, how many sessions they have, how many enquiries they make — these tell us way more about conversion likelihood than their stated budget or travel preferences. For profit, enquiry_touchpoints and page_interactions are by far the strongest signals (0.61 and 0.57). This makes intuitive sense people who engage more tend to book bigger, more customized trips.

Other observations:

- Paid social (mostly Facebook/Instagram) brings in about 30% of all leads, followed by direct traffic at 24%.
- 57% of leads come from mobile devices, which makes sense for a consumer travel product.
- The US accounts for 38% of leads. Australia and Canada are next at around 7% each.

- Most leads report a 'low' or 'not sure' budget, meaning they are still exploring.



5. Final Dataset Summary

	Details
Total rows	2,458 unique leads
Total columns	37 (1 ID + 34 features + 2 targets)
Numeric features	19
Categorical features 15	15
Target 1	target_conv_prob — conversion probability (0 to 0.66)
Target 2	target_profit — predicted profit if converted (26K to 670K)
Missing values	187 removed from original 224
Dropped columns	1,473 duplicates removed

The dataset is now clean and ready for the next phase.

😊 **END** 😊